

Build the partner around the factory

AIKYANTRA / XELOR opportunity and execution plan | 10 September 2026

RECOMMENDATION

Make AIKYANTRA the accountable technology partner. Use XELOR for intelligence and coordinated action, ONYX where a new operational record system is needed, and qualified specialists for equipment, integration and security.

The strongest opportunity is to connect a customer promise to materials, production, quality, delivery and money. The reviewed suppliers each cover useful parts of this chain. AIKYANTRA can combine these into a practical service for a defined manufacturing segment, then expand after proving results.

Three priorities

- Close the commercial gap: turn an RFQ and its drawings into a checked cost sheet, approved quotation and traceable job.
- Make the existing factory foundation dependable: real input data, simple operator workflows, scans, quality decisions and maintenance closure.
- Join purchasing to cash: supplier commitments, receipt/invoice matching, payable dates and an explainable cash view.

What this report is based on

All 23 supplied photos and both videos; nine supplier groups; official websites, public company social pages and QR/link checks; and a read-only audit of the current XELOR workspace. The companion research PDF gives one page per supplier. This report turns that evidence into recommendations.

Working assumption: the first target is a small or midsize discrete manufacturer, such as precision components, job work or industrial equipment. Team capacity, customer segment and budget were not specified. Priorities and timelines below are proposals to validate with a pilot, not delivery commitments.

Reading route	Pages
Current product and supplier comparison	2-5
Priorities and detailed capability additions	6-16
Partner delivery, packages and market focus	17-19
Execution plan, value measurement and acceptance	20-23
Evidence references and plain-language glossary	24

Evidence: supplied media; company research PDF; repository audit at commit 13e8ecf, branch xelor-phase-2. No production deployment or customer outcome was tested.

Start from what XELOR actually has

A source-level baseline, with a clear boundary around demonstrations

The current README separates ONYX, the ERP and system of record, from XELOR, the intelligence layer. Older documents sometimes call the combined product XELOR. In this report, "current foundation" means code found in this workspace, including the business workflows used by the intelligence layer.

Status	Evidence and implication
Implemented in source	Customer/orders, BOMs, stock ledger, purchase/receipts, production operations, inspections/CAPA, maintenance, MRP, heuristic finite scheduling, AR/accounting, approvals and mission evidence. These are useful foundations; production readiness was not established.
Real but bounded actions	Fulfilment has internal purchase-order and work-order writers and rereads their results. This is stronger than a suggestion-only demo, but is not proof of arbitrary external execution.
Demonstration / mock	Factory Intelligence computes OEE, yet its current scenario is configured mockOnly. Factory Connect includes simulated devices and catalogue targets. Working Capital uses hardcoded example figures.
Service demonstration	RELAY describes managed-service lifecycle and responsibilities using illustrative data. ACHILES has read-only health probes. Neither proves a staffed service or customer SLA.
Planned / not found	Full RFQ/CPQ, engineering change lifecycle, robust mobile/offline scanning, validated prediction, production third-party ERP/device connectors, complete direct-material AP and open marketplace operations.

Corrections that matter

- Do not rebuild all scheduling: a finite heuristic scheduler already exists. Extend and validate it.
- Do not call all quality "missing": inspections and CAPA exist. Deepen execution, measurement and genealogy.
- Do not advertise all 24 UI modules as 24 finished products. Module count is a navigation measure.

PRODUCT RULE

Mark every customer-facing workflow as live, limited, simulated or planned. Remove sample results from a paid pilot's performance claims.

Internal: X1-X8 (reference key on page 24). Read-only inspection; no historical test counts were rerun.

Preserve the foundation; complete the gaps

The next addition should close an entire customer workflow

Area	Current foundation	Useful next layer
Enquiry to order	Customers, orders, credit/tax checks	RFQ documents, cost model, quote versions, commercial approval, accepted-quote handoff
Engineering	Items, versioned BOM, basic where-used	Drawing revision, ECO impact, effective dates and acknowledgement at the workstation
Plan and produce	MRP, finite heuristic scheduling, operations	Validated constraints, live feedback, operator job tickets and controlled rework
Materials	Stock ledger, batches, warehouse types	Bins, scan validation, kit readiness, cycle counting and picking/dispatch tasks
Quality	Inspection specifications, NCR/CAPA	Gauge status, in-process capture, supplier loop, genealogy and recall rehearsal
Maintenance	PM, work orders, parts and downtime	Technician mobile evidence, condition inputs and verified alert response
Finance	AR, receipts, accounting, expenses	Supplier bills/three-way match, real AP, commitments and cash forecast
Connectivity	Import/retry patterns and ONYX HTTP port	One supported pilot connector, durable edge buffer, deduplication and reconciliation
Safety and security	App permissions, audit, safety flags	EHS workflows and qualified OT-security service; vision only after validation
Delivery model	Service and partner concepts	Named people, support coverage, commissioning packs and monthly outcome reviews

CHOICE

For an existing ERP customer, integrate the missing operational layer. For a plant without dependable records, deploy the necessary ONYX workflows and establish record ownership first.

A supplier's brochure is evidence of advertised scope. Repository code is evidence of implementation. Neither alone proves a reliable customer deployment. Compare these evidence types openly and require pilot acceptance before claiming equivalence.

Internal: X1-X10. Supplier scope: all nine profiles in the companion PDF.

What to learn from each supplier

All nine supplier groups mapped to an AIKYANTRA decision

Supplier / reference	Valuable pattern	Recommended response
CommerceCX / ScaleFluidly	RFQ-cost-quote-order continuity; loading optimisation; governed NPI	Build supervised RFQ/costing first. Integrate loading for frequent shippers. Reuse stage gates; defer clinical workflows.
AnjX Solutions	Forecast, inventory and capacity scenarios for SMEs	Improve planning scenarios and measured backtesting. Evaluate specialist algorithms only after real data is ready.
Slooze	Procurement decisions, supplier visibility and role-specific guidance	Build supplier RFQ/bid comparison, commitments and exceptions tied to jobs and budgets.
Tomax / D3Minds	Connected MES, warehouse, quality and engineering projects	Deepen factory execution and traceability. Use as a functional benchmark for demonstrations.
Zeliot / Condense	Reusable real-time data foundation and application delivery	Evaluate as an integration/platform option where workload economics justify it. Own factory workflows.
Adventis / OptiMES	Small operations modules plus gateways and maintenance/safety	Use modular packaging. Pilot a gateway/condition workflow with defined compatibility and support.
Controlsoft	Physical engineering, panels, commissioning and maintenance	Qualify for an installation/integration partner shortlist. AIKYANTRA retains customer accountability.
RheinBrücke / Epicor	ERP extensions plus full implementation and ongoing support	Learn delivery discipline and extend existing ERP. Evaluate partner route for larger ERP programmes.
Connectivity / CoSol + Cisco	Industrial connectivity, security and managed infrastructure	Partner for network/OT security and remote access. Keep this distinct from application security.

These are research-based options, not endorsements or confirmed partnerships. Check customer fit, delivery region, API access, licence economics, data ownership, support and reference projects before selection.

Primary links: [C2/C5](#); [A1/A2](#); [S1/S2](#); [T2/T3](#); [Z1/Z2](#); [B1/B3](#); [B9/B11](#); [B13/B15](#); [K1/K2](#). Link key on page 24 and companion profiles.

Design one connected customer journey

Commercial promise -> factory execution -> verified financial outcome



Each step should carry the same job identity, revision, required quantity, due date, responsible person and source evidence. A late supplier delivery must be visible to the planner, supervisor and customer-service owner without separate spreadsheets.

Layer	AIKYANTRA responsibility
Customer workspace	Owner, buyer, planner, operator, inspector and technician see the work relevant to their role. Shared facts; different actions.
XELOR intelligence	Detect exceptions, retrieve evidence, compare options, request approval, execute only supported actions and verify the result.
Operational workflows	ONYX or the customer's existing ERP owns agreed records. Quality, maintenance and warehouse actions create traceable transactions.
Data and connectivity	Versioned integrations join business records and factory events. Preserve source identity, timestamps, units and quality/freshness status.
Plant and infrastructure	Machines, sensors, meters, networks and security remain under documented local control, with qualified installers and support.
Delivery and service	Discovery, commissioning, training, incident response and continuous improvement run across all layers.

EXAMPLE

A bearing delivery slips. XELOR identifies affected jobs, checks available stock and a qualified alternative, estimates delivery/margin impact, obtains approval and updates the relevant purchasing/planning workflow. It rereads the result and records unresolved work.

The intelligence layer must not create a second uncontrolled stock or financial ledger. Data ownership and reconciliation are part of the product design.

Synthesis from C2, T2/T3, Z1, B13/B15 and X1-X9; the complete illustrated future journey is a recommendation.

Prioritise by value and readiness

Complete a repeatable workflow before expanding the catalogue

Priority	Deliverable	Why now / dependency
P0: essential	Pilot scope, accurate masters, supported connector/import, roles, backup/restore, real service owner	Every other promise depends on trustworthy data and a recoverable service.
P1: first package	RFQ intake, checked costing, quote approval and accepted-order link	Closes a clear revenue workflow gap and can start before machine connectivity. Needs commercial rules and real RFQs.
P1: alternative first package	Operator capture, one-line visibility, material scans and quality handoff	Choose instead when downtime, missing WIP or wrong material is the buyer's main pain. Needs site access and supervision.
P2: strengthen	Supplier commitments, AP connection, engineering change, traceability and maintenance closure	Improves delivery reliability and actual cost once the first records are dependable.
P3: selective	Forecast/APS depth, energy, loading optimisation, vision or predictive maintenance	Only where the customer has the problem, usable data and a measurable benefit.
Later	Open marketplace, broad digital twins, full clinical platforms, autonomous physical control	High scope and validation burden; weak reason to put them in a general MSME starting package.

How to choose between the first two packages

- High-mix engineer-to-order plant: quotation effort, drawing revisions and margin leakage usually make RFQ a better starting hypothesis. Validate in discovery.
- Repetitive production plant: poorly recorded stops, scrap and replenishment may make factory execution the stronger starting hypothesis.
- Use customer interviews and a baseline to choose one. Do not attempt both complete packages in a single undersized pilot.

Indicative effort bands: small means a contained workflow extension; medium means new records plus UI and integrations; large means site hardware, multiple systems or specialised validation. Estimate actual person-weeks after reviewing the pilot data and code.

Recommendation based on X1-X10 and the reviewed offerings; not a quantified demand survey.

Add RFQ-to-quotation continuity

Priority P1 | Owner: commercial lead + engineering | Effort: medium to large

Smallest useful version

- Receive the customer RFQ and attachments in one record. Record due date, customer revision, parts, quantities, delivery expectations and open questions.
- Extract text/tables with source-page links and confidence. A person confirms engineering requirements, units, quantities and ambiguous drawing details.
- Build a cost sheet from materials, cycle/setup time, labour, tooling, outsourced work, freight and overhead assumptions. Store supplier-price date and validity.
- Compare supplier bids on total comparable cost, quantity, lead time, tooling and payment terms. Keep assumptions visible.
- Apply a margin floor and approval limits. Issue a versioned quotation with validity, exclusions and delivery assumptions; link acceptance to the sales order.

Extend XELOR's existing work

Reuse customers, item/BOM records, workflow approvals and fulfilment mission evidence. Add RFQ, quote, quote revision and cost-assumption entities rather than placing financial promises only in chat. Link accepted quote lines to job/stock/quality records so actual cost can be compared later.

EXAMPLE

A drawing revision adds a machining step after a quote was drafted. Show the changed requirement, affected cost, revised lead time and required approval before a new quote is issued.

Acceptance and value

Use a representative set of real RFQs, including revisions and missing data. Trace every critical extracted value to a source or human correction. Prove that outdated quotes cannot be silently overwritten and acceptance creates the correct order only once. Measure median and slow-tail quote turnaround, correction rate and estimated-versus-actual job margin.

Defer autonomous geometric interpretation, unsupported win-probability scores and auto-issued commercial commitments. AI can prepare the work; authorised people own the engineering and price decision.

Connect procurement to production and cash

Priority P2 | Owner: purchase + finance | Effort: medium to large

What to add

- Maintain approved suppliers, capabilities, documents, lead times, contact roles and review/expiry dates. Link supplier performance to actual receipts and defects.
- Generate purchase RFQs from job/material demand. Normalise bid units and quantity breaks; compare delivery, payment, tooling, transport and quality terms.
- Track PO acknowledgement, promised delivery and changes. Escalate shortages against the affected production jobs, not just a generic late-PO list.
- Complete supplier-bill processing or connect the accounting system: match PO, receipt and invoice; route price/quantity exceptions to an authorised person.
- Build an AP schedule and a 13-week cash view from real due dates, committed purchases and expected receipts. Separate confirmed entries from scenarios.

Build on existing code

Purchase orders, GRNs, approval patterns and stock movements already exist. Direct-material AP/three-way matching needs completion or a reliable accounting integration. Working Capital's attractive sample figures must be replaced with reconciled calculations before they are used in a customer decision.

VALUE

The buyer sees which material will delay a job; the owner sees how an alternate supplier affects margin and cash; accounts sees the approved amount and due date. All three views use the same evidence.

Acceptance

Demonstrate partial receipts, price variance, duplicate invoice, rejected stock and credit adjustment. Reconcile committed quantities and amounts with the source ledger. Measure supplier on-time delivery, receipt-to-bill effort, exception backlog and forecast error. Payment execution remains outside the pilot unless specifically scoped and controlled.

Finance or payment products should be integrated through qualified providers if later needed. A procurement screen does not make AIKYANTRA a financing provider.

Supplier references: Slooze [S1/S2](#) and CommerceCX image 13. Internal: X2/X8/X9.

Improve planning without replacing what works

Priority P2-P3 | Owner: planner | Effort: medium to large

XELOR's foundation already includes MRP and finite heuristic scheduling. The valuable addition is a plan that matches plant constraints, explains conflicts and responds to reliable actual progress.

Step	Required improvement
Master the constraints	Verify calendars, actual capacity, qualified alternate machines, routing, setup time, tool/fixture availability, skills and subcontract lead times.
Make demand explicit	Keep customer orders separate from forecast. Identify assumptions and avoid counting the same demand twice.
Test practical scenarios	Compare overtime, alternate machine, split batch, subcontract and changed priority. Show cost, material and delivery consequences.
Publish with control	Planner reviews feasibility and uncertainty. Record the schedule version, freeze window and authorised override.
Learn from execution	Feed accepted output, stops, rework and material arrivals into the next planning cycle. Record why actual time differs from planned time.

Where AnjX is especially useful as a reference

Its forecasting, inventory and scenario approach suggests a useful next layer for repeat demand and spare parts. Start with simple, explainable forecasting benchmarks. Evaluate more advanced models against historical holdout periods and service/cash outcomes, not only a headline accuracy number.

VALIDATION

A proposed plan must not allocate a machine, tool or material twice. Test a breakdown, absent operator, late material and urgent order. The planner must understand why a job moved.

Track on-time-in-full delivery, schedule adherence, planning effort and expedite cost. Forecast evaluation should state horizon, item mix, data exclusions and error measure. A well-tuned planning rule can be more useful than a model trained on unreliable history.

Turn factory visibility into useful daily work

Priority P1 alternative | Owner: production supervisor | Effort: medium + site work

One line, one daily routine

- Show the operator the released job, correct drawing/work instruction, quantity and inspection requirement. Support start, pause, complete, reject and reason capture.
- Start with manual counts and a small number of validated signals. Connect machine counters only after units, resets, timestamps and job assignment are understood.
- Give the supervisor a shift board: target, accepted output, stops, top losses and unresolved actions. Capture the owner and due time of each response.
- Link good output to finished/WIP stock; link rejected output to quality disposition; link downtime requiring repair to maintenance.

Make OEE honest

OEE = availability x performance x quality. Define planned production time, run time, ideal cycle and total/good count for each comparable process. Missing inputs produce an "insufficient data" state rather than an invented percentage. Reconcile counters and prevent overlapping downtime from being counted twice.

CURRENT GAP

OEE calculation exists in source, but the current Factory Intelligence scenario is explicitly a mock. The first deliverable is dependable input and operational use, not another OEE dashboard.

Acceptance

Prove a complete shift: start job, pause for an explained stop, record a reject, complete inspection and post accepted output. Simulate a lost connection and counter reset. Reconcile the shift board to work-order and stock totals. Agree signal freshness and data coverage with the customer.

Measure unplanned-stop hours, accepted output, scrap, manual reporting effort and unresolved actions. Compare similar product/shift conditions. Higher measured downtime after better capture may reveal old losses; it is not automatically worse performance.

Join engineering changes, quality and traceability

Priority P2 | Owner: engineering + quality | Effort: medium to large

Engineering control

Add a drawing/document lifecycle and engineering change request/order. Show affected BOMs, quotations, open orders, bought material and work in progress. Define the effective revision and capture shop-floor acknowledgement. Preserve old records when a new revision is released.

Quality at the point of work

- Show the correct inspection plan for that operation and revision. Capture measurement, gauge, operator, time, lot and disposition.
- Warn or block use of an overdue gauge according to the customer's quality rules. Add calibration records and evidence.
- Open nonconformance from failed checks; contain affected stock; assign action and verify effectiveness through the existing CAPA foundation.
- Connect supplier defects, internal rework and customer complaints. Add an 8D-style customer response when the target segment requires it.

Trace the product in both directions

Link supplier heat/lot to receipt, component issue, operation, inspection, finished serial/lot and customer shipment. Keep rework, splits, merges and scrap visible. Existing batch/ledger records are the starting point; a full recall graph is still an addition.

ACCEPTANCE EXERCISE

Choose a suspect supplier lot. Identify stock still on hand, all affected jobs and shipped lots, relevant inspection evidence and the customer list. Rehearse containment without corrupting the ledger.

Measure first-pass yield, repeat defects, rework hours and trace-query completeness/time. Add SPC only after stable definitions and measurement quality exist. APQP/PPAP, full PLM and regulated validation are segment-specific extensions, not automatic parts of every MSME package.

CommerceCX's pharma material offers useful patterns for approval gates and records. Full clinical research and cell/gene therapy operations would require a separate product strategy and specialist validation.

Make warehouse and dispatch traceable

Priority P1-P2 | Owner: stores + logistics | Effort: medium

Workflow	Smallest valuable addition
Receive and locate	Scan item/lot, validate PO and quantity, capture quality hold, print identifier and choose bin/location.
Count and replenish	Cycle counts with controlled adjustments; low-stock and production-line replenishment tasks; clear ownership.
Kit and issue	Reserve material to a job, check correct revision/lot, show missing components and record the issue once.
Handle WIP/rework	Retain lot identity through partial completion, movement, rework, scrap and return.
Pick, pack and dispatch	Validate customer/order/item/lot/quantity; produce packing labels and dispatch evidence; update the order and inventory consistently.

Start with scans people can use

Use a phone/tablet or existing scanner where suitable. Test labels under actual lighting, dust and handling. A scan should populate a transaction with clear confirmation and error messages, not merely open a generic webpage.

Where loading optimisation fits

CommerceCX LogisticCX is relevant when recurring container or truck loads create meaningful cost. First capture dimensions, weight, quantity, orientation, stackability, unloading order and loading constraints. A 3D picture alone does not prove a safe or executable load.

ACCEPTANCE

Use a wrong lot, duplicate scan, damaged label, partial pick and network interruption. Verify that no duplicate stock movement occurs and all exceptions are resolved with an audit record.

Measure inventory accuracy, picking errors, kit-related stoppages and dispatch delay. For a loading pilot, compare the proposed plan with actual execution and agreed freight/labour results. Defer forklift optimisation and complex robotics until volume and infrastructure justify them.

After delivery: service, returns and spares

For equipment makers, extend the shipped serial/lot record into warranty, service requests, approved returns and spare-parts demand. Connect recurring field failures to quality and engineering changes. Begin with complaint/return ownership and traceability; add a customer portal or field-service scheduling when volume justifies it.

Supplier: Tomax supplied image 21; CommerceCX C6 and image 16. Internal: X4/X6/X9.

Close maintenance work; then add prediction and energy

Priority P2, advanced analytics P3 | Owner: maintenance | Effort: medium + hardware

Make the maintenance foundation operational

- Clean the asset register, failure codes, criticality, preventive tasks and spare-parts mapping. Record downtime against an asset and affected operation.
- Give technicians a mobile work order with checklist, isolation/permit reference, photos, readings, parts used and completion evidence.
- Convert a useful condition alert into an assigned work request with priority and due date. Close it only with a recorded inspection/action result.
- Review recurring faults, overdue PM, repeat failures, parts shortages and repair time with the production supervisor.

Earn the predictive claim

Add vibration, temperature, current or other signals only for a defined failure mechanism and asset group. Partner for sensors and interpretation. Establish a labelled history and validation method. Measure useful warning lead time, false alarms, missed events and whether the team acted. Until then, call the feature condition monitoring or threshold alerts.

Energy and utilities can be a separate small pilot

Meter the relevant equipment or line. Identify idle-load waste and unusual use; calculate energy per accepted unit when job attribution is reliable. Start with electricity or compressed air where the customer already suspects losses. Add water/effluent workflows only for a relevant process plant.

ACCEPTANCE

Demonstrate alert -> owned work -> parts/action -> verified closure -> downtime record. For energy, reconcile meter totals and state the allocation method before assigning cost to a job.

Track planned-maintenance completion, repeat failures, unplanned downtime and maintenance effort. Distinguish avoided cost from extra saleable output. Existing software PM and KPI functions should be hardened rather than recreated.

Treat factory safety and security as distinct workstreams

Priority P0 for basic controls; specialist extensions by customer need

Area	What AIKYANTRA should provide	Specialist boundary
Application security	Roles, least privilege, approval controls, audit, tenant separation, tested recovery and change handling	Validate actual deployment; source code controls alone are not a security audit.
OT network security	Maintain an asset/connection register and approved access/change records in the customer workspace	Partner for passive discovery, network segmentation, industrial monitoring and secure vendor access.
EHS workflows	Hazard and incident records, near misses, permits/checklists, actions, training evidence and review dates	Plant safety owner approves the process and required controls.
Vision alerts	If justified, route camera detections to trained human review and track outcomes	Validate lighting, PPE types, occlusion, false alarms and missed detections with a specialist.
Remote support	Named access request, approval, time limit, purpose and service ticket; record closure	Security partner configures secure access and revocation. Local plant authority remains explicit.

Connectivity/CoSol and Cisco show the gap between securing a web application and securing factory networks. Adventis shows how EHS can share asset and work records with maintenance. These are complementary capabilities, not interchangeable certifications.

OPERATING BOUNDARY

The first factory integration should collect data and support human decisions. Autonomous machine actuation and emergency protection are separate engineering scopes with explicit site/OEM approval and validation.

Acceptance

Review a vendor-access session end to end, including expiry/revocation and an incident escalation. Rehearse restoration from backup. For EHS, trace an incident through assigned action and review. Never describe a camera or software installation as guaranteeing a hazard-free or certified plant.

Primary: Cisco Cyber Vision / Secure Equipment Access (K2/K3); Connectivity K1; Adventis B1. Internal: X5/X10.

Productise integration and data quality

Priority P0 | Owner: integration lead | Effort: medium to large

An MSME will judge the system by whether it agrees with the shop floor and accounting records. A reusable, supported integration is a product in its own right.

Requirement	Practical implementation
Record ownership	For each customer, item, BOM, order, stock and invoice, name the authoritative system and allowed write path.
Versioned mapping	Record external IDs, units, currencies, time zones and revision rules. Validate inputs and quarantine uncertain records.
Reliable delivery	Use durable queues/buffers, explicit acknowledgements, retries, idempotency keys and duplicate protection.
Reconciliation	Compare source and destination counts, quantities and values; assign exceptions with a retry/review trail.
Device resilience	Handle reconnects, counter resets, missing timestamps and stale signals. Store source and ingest times separately.
Supportability	Show connector health, last successful sync, backlog and failure owner. Provide controlled replay and rollback procedures.

Choose a narrow first connector

Use the actual pilot's ERP/accounting version and machine/controller. A checked file import may be a sensible start when real-time exchange is not needed. Do not promise universal SAP, Tally, Odoo or Dynamics compatibility because an integration catalogue contains a name.

Where Zeliot fits

Condense is worth evaluating when live data, many integrations and ongoing platform operations become material. Compare cloud, platform licence, engineering and support cost for the actual workload. Avoid imposing enterprise streaming economics on a small one-line pilot.

RELEASE GATE

Prove repeated import, partial failure, duplicate delivery, outage recovery, stale data and reconciliation on the customer's real data. Only then label the connector supported.

Supplier: Zeliot [Z1/Z5](#); Adventis [B4](#); RheinBrücke [B21](#). Internal: X5/X9.

Give workers simple tools and AI a clear job

Priority P1-P2 | Owner: product + plant champion | Effort: medium

Worker experience

- Build touch-friendly task flows: start/finish a job, scan material, record inspection, raise a fault and complete maintenance. Use the plant's working languages where discovery shows need.
- Show only the current task, correct instructions and required evidence. Design for gloves, shared devices, noisy areas and short training sessions.
- Support offline capture only with a durable queue, clear pending/synced status and supervised conflict resolution. A responsive website is not automatically an offline application.
- Start with a tested web/PWA approach if it meets the site needs. Build a native app only when device integration, resilience or deployment requirements justify it.

Useful AI tasks

Task	Required behaviour
Read and prepare	Draft RFQ fields, supplier comparison or maintenance summary with sources and uncertainty.
Explain an exception	Identify the evidence behind a late job, shortage, defect or cash risk; distinguish known facts from inference.
Compare options	Calculate consequences using governed rules and current data. Explain missing inputs.
Act within authority	Use an existing supported executor, obtain required approval, reread the result and record an outcome.
Know when to stop	Escalate when data is stale, permissions fail or the action is unsupported. Preserve a usable manual path.

The existing agent governance, evidence and approval patterns are an advantage to preserve. Add useful capabilities to that common structure instead of creating separate ungoverned chatbots for every module.

SUCCESS TEST

An operator can finish the core task reliably after practical training, and a manager can explain every AI-assisted business action from its records.

Supplier: CommerceCX image 13; Adventis image 10; Tomax image 21. Internal: X7/X9 and operator-workflow source inspection.

Make end-to-end delivery a real service

AIKYANTRA owns the outcome even when specialists do part of the work

Stage	Named accountability	Customer deliverable
Discover	Solution owner + plant sponsor	Process walk, systems/assets, pain baseline and fixed pilot scope
Design	Solution owner + domain/integration leads	Workflow, record ownership, hardware list, access plan and acceptance criteria
Implement	Delivery lead; qualified installer for site work	Configured workflows, clean masters, connected devices, reconciled data and commissioning evidence
Train and stabilise	Plant champion + customer-success lead	Role training, open-issue log, fallback procedure and acceptance sign-off
Operate	Service owner + contracted support partners	Coverage calendar, ticket triage, monitoring, recovery and escalation
Improve	Account owner + plant sponsor	Monthly outcomes, support cost, adoption review and next justified improvement

What to build, buy and partner for

- Build/own: common workflow and evidence model, approvals, manufacturing decision logic, connector contracts, customer experience and delivery method.
- Buy/integrate: appropriate accounting/ERP, commodity OCR, cloud, identity, monitoring and ticketing. Do not recreate mature tools without a customer reason.
- Partner: PLC/SCADA/DCS engineering, electrical panels, meters/sensors, OT security, specialist vision, robotics and regulated validation.

PARTNER AGREEMENT

Specify region, skills, lead time, acceptance tests, response coverage, escalation, warranty, access handling and who pays for rework. Give the customer one AIKYANTRA delivery owner.

RELAY can become the operating workspace, but its present demonstration is not a staffed service. Begin with a curated partner and connector catalogue. Add an open marketplace only when installation quality, entitlement, support and commercial settlement can be operated consistently.

Supplier references: Controlsoft [B9/B11](#), RheinBrücke [B13](#), Connectivity [K1](#). Internal: X10.

Sell understandable packages with visible cost

Commercial design hypotheses, not researched vendor price quotes

Package	Customer promise	Scope and success measure
Assess	Know which problem to fix first	Site/data/process review; baseline; scoped business case and pilot acceptance plan
Quote to Job	Prepare and approve a dependable quote faster	RFQ, checked costing, revision/approval and accepted-order link; measure turnaround and corrections
Factory Execution	Know what is happening and close the action	One line, operator input, selected signals, quality/material handoff; measure coverage, losses and closure
Cash and Control	See supplier commitments and cash consequences	Receipt/invoice exception flow, accounting connection and forecast; measure reconciliation and error
Operate and Improve	Keep the service useful after go-live	Named support coverage, connector monitoring, restore drills, training and monthly outcome review

Show the full cost

Separate diagnostic/pilot, implementation, hardware/installation, software/site subscription, connectors, cloud/AI usage and support. State inclusions, volume assumptions, travel, third-party licences and change requests. Give customers data export and an agreed exit/handover process.

Set price from delivery economics and customer value

One-time cost = discovery + configuration + data cleanup + integration + installation + training. Monthly cost = hosting/licences + support labour + monitoring + partner coverage + model usage. Test the price against credible benefit and willingness to pay; do not set it by copying unverified competitor percentages.

SCOPE RULE

A small module can have substantial installation or support cost. Price the complete working outcome and put a boundary around custom development.

Public monetary pricing was generally unavailable. Zeliot's calculator separates platform and cloud cost, so its advertised entry figure is not an all-in factory deployment quote. No vendor-price ranking is justified by this research.

Commercial synthesis; public model evidence: Tomax T1, Zeliot Z5 and reviewed supplier pages.

Choose one entry segment; adapt the package

A practical starting hypothesis for manufacturing MSMEs

Plant type	Likely first problem to test	Product emphasis
Precision / job-work components	RFQ effort, revisions, due dates and lot history	Quote to Job; routing/cost assumptions; ECO; basic traceability
Pumps / machinery / engineer-to-order	BOM changes, purchased parts, project dependencies	Engineering/project gates; procurement commitments; job margin; delivery risk
Repetitive parts / assembly	Stoppages, replenishment, scrap and shift reporting	Factory Execution; OEE with valid inputs; kitting; in-process quality
Export-heavy manufacturer	Dispatch preparation and recurring load cost	Traceable packing and dispatch first; constrained load optimisation where justified
Process / food / chemical	Batch process, utilities, hazards and process quality	Separate discovery and specialist design; recipe/batch, EHS and instrument needs change scope
Regulated pharma / clinical	Validation and domain-specific records	Partner-led specialist offering; do not extend generic MSME claims into clinical operations

Recommended starting point

Pick one accessible discrete-manufacturing plant with a committed owner, a daily user champion and a measurable bottleneck. A complex but observable workflow is more useful than an impressive demo with no accountable user.

Qualify the pilot

- Can the customer provide representative records and permit process observation?
- Can a named person approve master data, costs, quality rules and acceptance?
- Is there one workflow whose benefit can be measured within the pilot period?
- Can the plant operate a fallback if the new workflow or connector is unavailable?

This segment order is a strategic recommendation, not a market-size study. Confirm willingness to pay, competition, sales cycle and field-service reach before choosing a broader market.

Inference from supplier scope and XELOR baseline; no TAM or customer-demand survey conducted.

Run a focused 90-day pilot

Sequence is conditional on team, access, data and technical readiness

Period	Work	Exit evidence
Days 1-15	Select one plant and one outcome. Observe work, audit data, inventory simulated features, agree scope and baseline.	Named sponsor/champion; data and asset register; pilot scope; system ownership; success and stop criteria
Days 16-30	Validate technical design and current code. Clean masters; configure roles; build needed import/connector and core task flow.	Reconciled test records; permissions; failure/recovery plan; demonstrable user workflow
Days 31-45	Complete exception paths, field setup if included, role training and shadow run.	Users perform representative cases; duplicates/outages handled; backup restore demonstrated
Days 46-75	Run bounded live work with authorised approvals. Review data quality, usage, failures and support effort weekly.	Timestamped records; reconciled totals; issue ownership; comparison periods and service labour
Days 76-90	Review customer benefit, unresolved issues and delivery economics. Document reusable setup.	Customer acceptance or corrective plan; measured outcome; support cost; expand/fix/stop decision

Keep the pilot narrow

For Quote to Job, cover a defined RFQ family and its order handoff. For Factory Execution, cover one line/area and a defined job/quality/material flow. Add a second package only if capacity and acceptance evidence support it.

STOP OR RESET

If core data cannot be reconciled, users cannot complete work, or no accountable customer owner exists, fix the prerequisite before adding AI or expanding to more machines.

Schedule changes should follow evidence. Hardware lead times, ERP access and source-code readiness can move these windows. The report does not assume every proposed module can be built in 90 days.

Execution recommendation built on the source audit and supplier delivery lessons.

Expand over a year through reusable delivery

Milestones are conditional; staffing shown as roles, not current headcount

Stage	Expansion	Release condition
First quarter	One accepted workflow at one pilot plant	Reliable user completion, reconciled data, recoverability and measured service cost
Second quarter	Repeat at another similar plant; add traceability/quality or procurement depth	Reuse configuration/connector with less delivery effort and no hidden data divergence
Third quarter	Add one advanced option: planning, energy, condition monitoring or loading	Customer problem and data justify it; specialist delivery and validation agreed
Fourth quarter	Broaden within the selected segment; strengthen managed support and partner catalogue	Service margin, support coverage, onboarding quality and customer retention justify expansion

Minimum responsibilities to cover

- Product/solution owner: customer workflow, priorities and acceptance.
- Manufacturing domain lead: engineering, planning, quality and operational definitions.
- Application engineers: backend/records, frontend/worker experience and meaningful validation.
- Integration/edge lead: ERP/device connection, data quality and recovery.
- Customer-success/service owner: training, adoption, incidents and outcome review.
- Qualified field/security partners: site installation, networks and specialist scope.

One person may cover multiple roles in an early team, but every role needs allocated time. Do not plan several independent modules as if the same engineer and domain lead can deliver all of them simultaneously.

REUSE TARGET

Build a repeatable deployment kit: discovery checklist, master templates, connector configuration, acceptance cases, training materials and service runbook. Track hours saved on the next installation.

Keep platform work tied to delivery: feature entitlements, configuration, audit, export, versioned upgrades and partner access. A marketplace becomes useful after the service catalogue is repeatable, not before.

Recommended operating plan; no current team capacity or delivery budget was supplied.

Measure value without double counting

Agree operational evidence and financial interpretation with the customer

Outcome	Measure	Evidence discipline
Quote speed	Median and 90th-percentile receipt-to-approved-quote time	Compare similar RFQ complexity; count missing-information pauses explicitly
Delivery	On-time-in-full shipments / due shipments	Freeze promised-date policy; do not move dates to improve the result
Factory reliability	Unplanned stop hours; valid event coverage; action closure	Use comparable operating time/product mix; exclude overlapping stops
Quality / material	First-pass yield, rework hours, stock agreement, picking errors	Use actual inspection, movement and physical-count records
Cash	Collections, slow-stock release, forecast error	Reconcile ledger/bank evidence; separate assumptions from actual cash
Service economics	Implementation hours and support hours/site/month	Include partner, hosting and issue-resolution cost, not only licence revenue

Illustrative example only - not a forecast

Suppose comparable RFQs fall from 3 hours to 1.5 hours each, with 40 RFQs a month. That releases 60 hours/month. At an assumed loaded cost of Rs 500/hour, the capacity value is Rs 30,000/month. It becomes a cash saving only if overtime, outsourcing or another actual cost falls; otherwise it is freed capacity.

Suppose validated reductions in scrap and overtime total Rs 35,000/month and recurring software/service cost is Rs 15,000/month. Net recurring benefit is Rs 20,000/month. An assumed Rs 180,000 setup cost then has a simple payback of 9 months, before tax, financing and other effects. Replace every input with customer evidence.

KEEP BENEFITS SEPARATE

Cash savings, released working capital, freed staff capacity and extra contribution from saleable output are different. Do not count the same recovered machine hour as all four.

No supplier's advertised percentage has been used as AIKYANTRA's expected return. Establish baseline, comparison window, data coverage, confounding changes and customer sign-off before publishing a case study.

Illustrative arithmetic authored for this report; supplier results remain vendor claims. Measurement recommendations draw on XELOR transaction/evidence design.

Use acceptance gates and targeted vendor checks

Turn broad promises into reviewable demonstrations

Gate	Evidence required before customer release
Workflow	Real users complete normal and exception cases with correct records, approvals and ownership.
Data	Required masters are approved; critical values trace to a source; source/destination totals reconcile.
Resilience	Repeated requests, duplicate scans, partial failures, reconnects and restore are tested. Pending work is visible.
Access and action	Permissions, tenant separation and approval limits hold in the deployment; supported writes are reread and recorded.
Adoption and service	Plant champion trained; fallback available; named service owner, coverage and escalation are operating.
Value	Baseline and measurement method agreed; results distinguish evidence, assumptions and actual benefit.

Supplier-specific demonstrations to request later

- CommerceCX: a real revised RFQ through checked costing, approval and ERP/order handoff; a physically constrained loading case if relevant.
- AnjX / Slooze: historical data with declared holdout/assumptions; shortage or supplier change reflected in material, schedule and cash decisions.
- Tomax / Adventis: one connected job across machine/operator input, inspection, maintenance and inventory; outage and false-alert behaviour.
- Zeliot: actual source protocols and schema changes, throughput/cost estimate, replay/recovery and ownership of custom connectors.
- Controlsoft / Connectivity: comparable commissioned project, local response, installation scope, access handling, acceptance and warranty.
- RheinBrücke: exact ERP/add-on licences, migration/integration scope, delivery effort, training, support and total cost.

Ask for shipped-versus-roadmap status, customer references, API/version documentation and product-specific security evidence. None of these supplier follow-ups were sent during this research.

DECISION

Expand only when the workflow works, people use it, records reconcile, service can support it and the customer accepts the outcome.

Derived from the observed brochure/public-evidence gaps and the source-level XELOR baseline.

Evidence key and plain-language glossary

Trace the recommendations back to suppliers and the current workspace

Research date: 10 September 2026. Repository: xelor-phase-2, commit 13e8ecf (6 September). Source code was inspected read-only; no live factory, external ERP deployment, model benchmark, security audit or customer outcome was verified.

Internal key	Representative source locations
X1 - positioning	README.md:1; apps/web/src/modules/registry.ts:46
X2 - commercial/fulfilment	apps/api/src/modules/sales/sales.service.ts:390; apps/api/src/fulfilment/fulfilment.module.ts:20
X3 - engineering/planning	apps/api/src/modules/engineering/engineering.service.ts:311; apps/api/src/modules/planning/schedule.service.ts:22; policy.service.ts:136
X4 - production/inventory	apps/api/src/modules/production/production.service.ts:415; apps/api/src/modules/inventory/inventory.service.ts:184
X5 - factory boundary	apps/api/src/agent-os/factory-intelligence.service.ts:477; packages/platform/src/factory-intelligence/oeo.ts:15; docs/01-agent-os/06-factory-connect.md:9
X6 - quality/maintenance	apps/api/src/modules/quality/qms-workflow.service.ts:47; apps/api/src/modules/maintenance/maintenance.controller.ts:184
X7 - agent controls	apps/api/src/agent-os/agent-control.service.ts:17; agent-authorization.service.ts:17
X8 - finance boundary	apps/web/src/modules/working-capital/workspace.tsx:30; apps/api/src/modules/accounts/accounts.service.ts:318
X9 - integration	apps/api/src/modules/dataimport/dataimport.service.ts:53; apps/api/src/modules/integration/message.service.ts:108
X10 - service boundary	apps/api/src/modules/managed-services/managed-services.service.ts:7; apps/api/src/modules/platform-health/platform-health.service.ts:39

Supplier IDs in strategy footnotes are clickable primary-source links. Their global mapping is saved as source_link_key.json. The companion PDF uses profile-local source labels. Full internal evidence and limitations are in xelor_baseline.md. This is a research and product assessment, not proof that any vendor or AIKYANTRA has delivered every advertised capability.

RFQ request for quotation; **CPQ** configure, price, quote; **BOM** bill of materials; **MRP** material requirements planning; **APS** advanced planning/scheduling; **MES** shop-floor execution software; **WMS** warehouse management; **QMS** quality management; **CAPA** corrective/preventive action; **CMMS** maintenance management; **OEE** availability x performance x quality; **EHS** environment, health and safety; **OT** operational technology; **IIoT** connected industrial devices; **AP/AR** supplier payables/customer receivables; **ECO** engineering change order; **SPC** statistical process control.

Authored synthesis and recommendations are explicitly separated from supplier claims and repository facts.